

CADD (2026)

Tutorial 1

Protein visualization using Chimera

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What is Chimera and why is it useful?

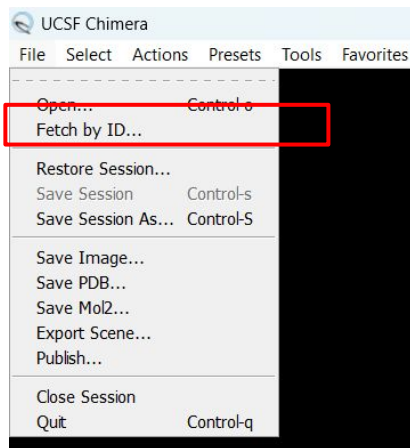
- Chimera is a **molecular visualization and analysis tool**
- Developed by UCSF for **structural biology and biophysics**
- Works with **experimentally determined structures** (X-ray, NMR, Cryo-EM)

Allows:

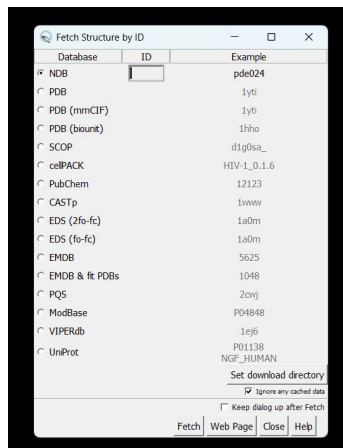
- Structural inspection
- Identification of **binding sites**
- Analysis of **noncovalent interactions**
- Comparison of structures (conformational changes)

1. Load the structure :

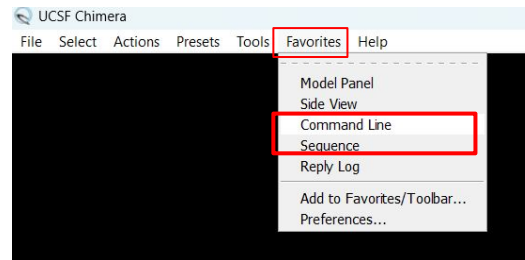
(i) Open downloaded Structure



(ii) Fetch by ID



(iii) Command line

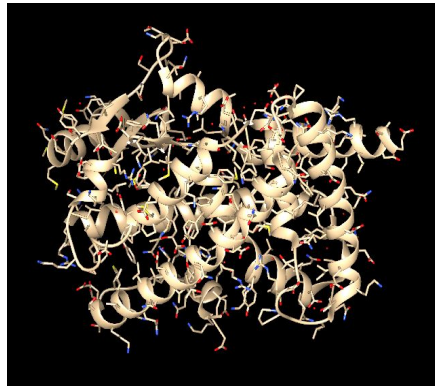
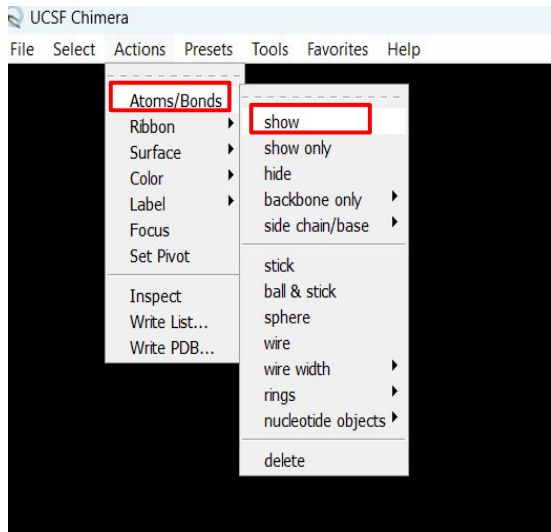


Command : open 1i7i

Link to the documentation:

<https://www.cgl.ucsf.edu/Outreach/Tutorials/GettingStarted.html> (UCSF CHIMERA)

2. Display atoms:



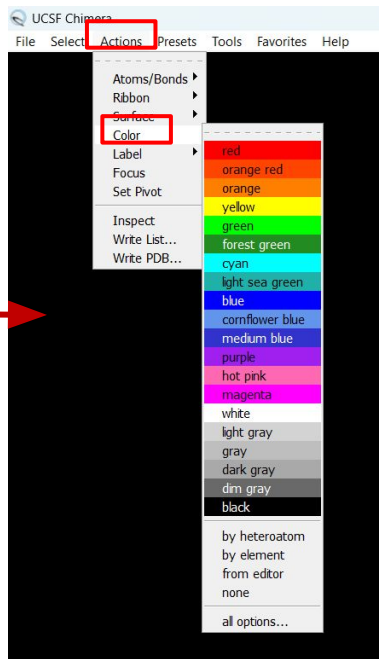
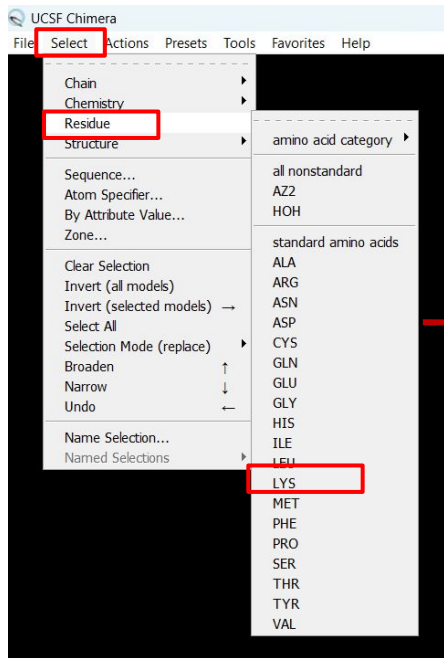
Color-coded by element:
oxygen red, nitrogens blue, etc.
The carbons retain the model
color, in this case tan.

Command : display

~display
(to undo display)

Command : ribbon
(for ribbon display)

3. Color :



Command: color green :lys

A blank specification is interpreted as all applicable items. For example, the following makes everything green:

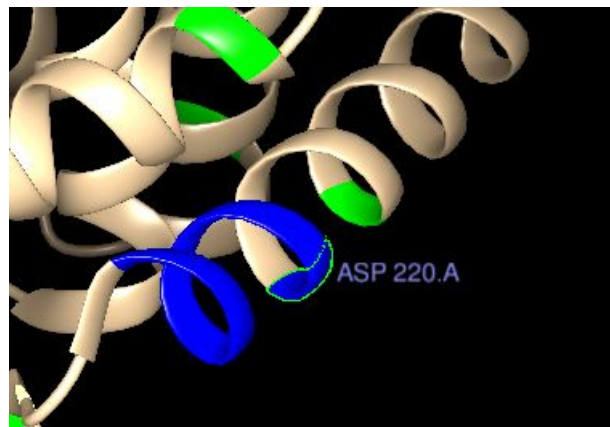
Command: color green

Command: color blue :220-225

4. Select:

Ctrl + click to start selection
Shift-Ctrl-click to add/subtract

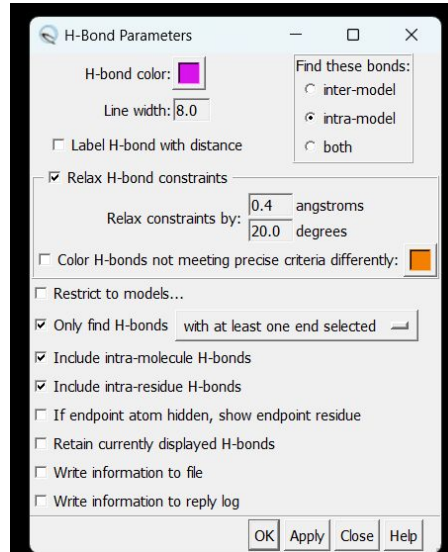
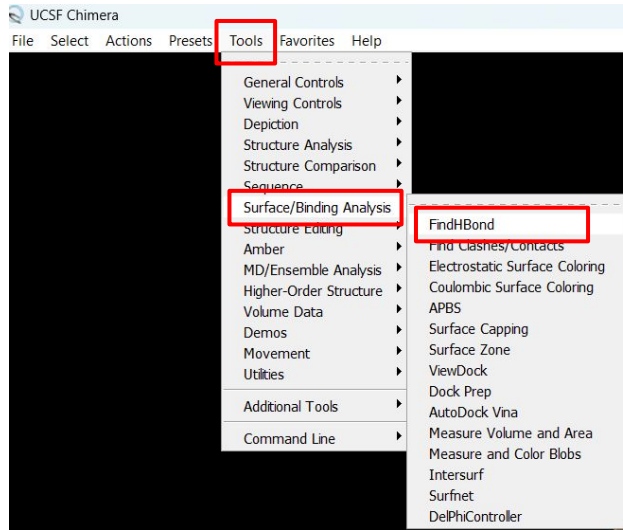
Command : sel :220
rlabel sel



Atom Specification Symbols		
Symbol	Meaning	Usage
#	model	# <i>model</i> (model ID number)
:	residue	: <i>residue</i> (residue name or number)
::	chain	:: <i>chain</i> (chain ID)
@	atom	@ <i>atom</i> (atom name)
=	partial wildcard	matches partial atom or residue name, e.g., @C= specifies all atoms with names beginning with C
?	single-character wildcard	matches single character in atom or residue name, e.g., :G?? specifies all residues with three-letter names beginning with G

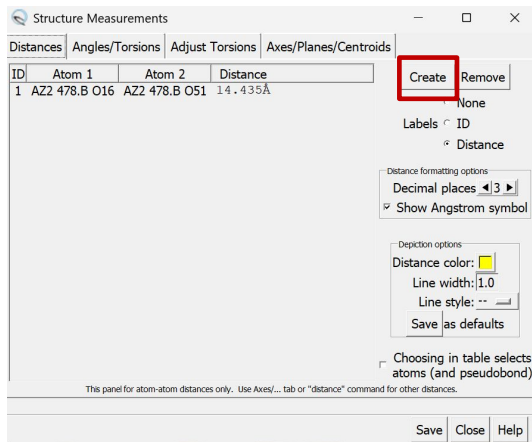
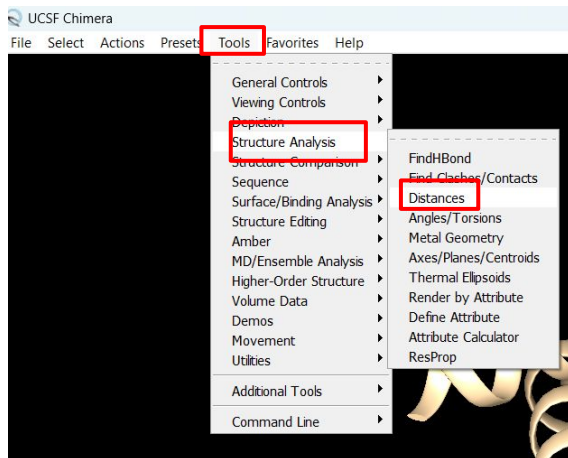
disp :220 z<8 - To display all the residues within 8Å distance of residue 220

5. Find Hbonds:

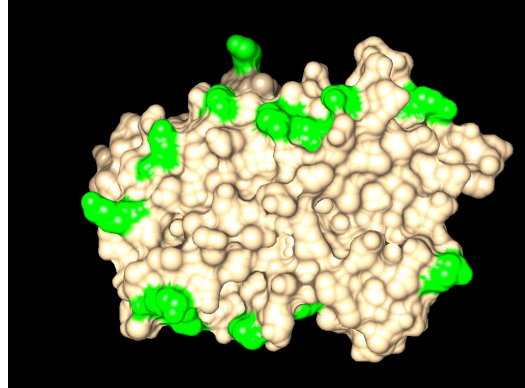
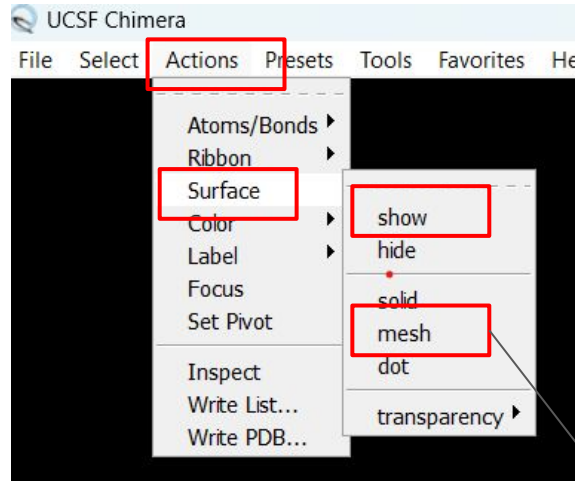


6. Distance:

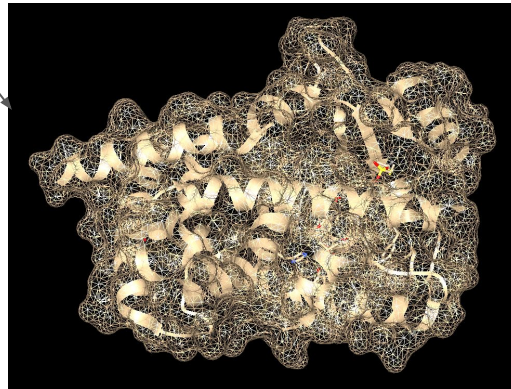
1. Select residues
2. Display their atom



7. Representation:



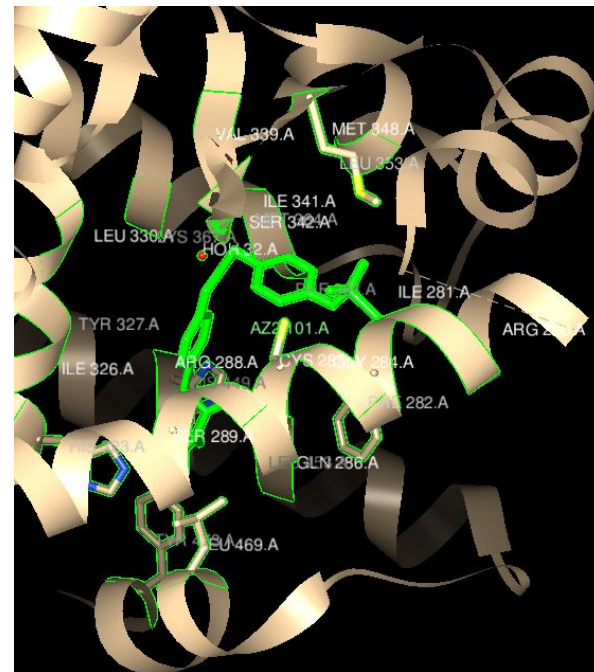
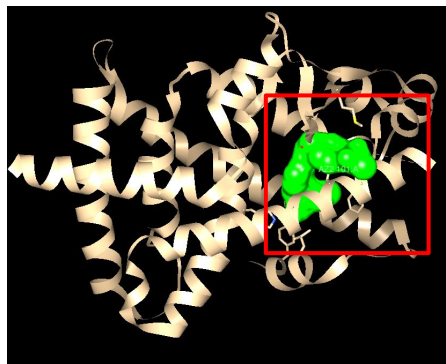
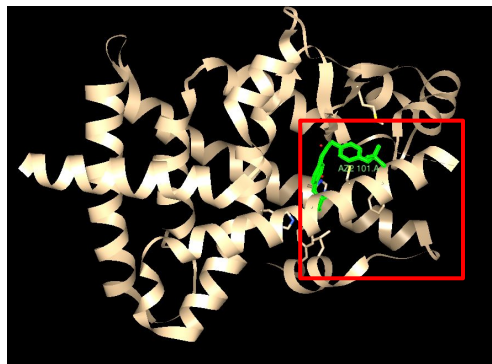
Command : surface
~surface



Command : surfrep mesh

8. Active site:

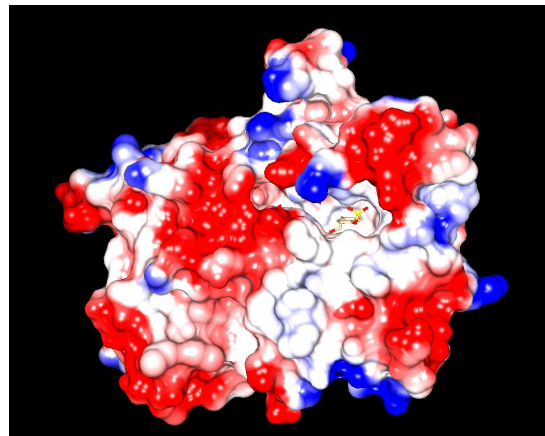
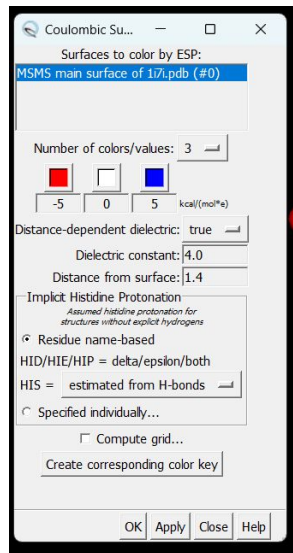
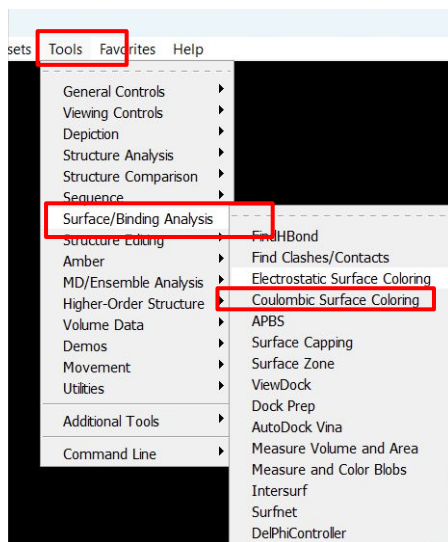
1. Select ligand
2. Color green sel
3. rlabel ligand
4. surface sel
5. focus sel (to zoom in)



select :AZ2 z<5.0 (To select all the residues within 5 Å range of the ligand)

rlabel sel

9. Coulombic representation



● **Red** → **negative electrostatic potential**

- Electron-rich regions
- **acidic residues** (Asp, Glu)
- attractive to **positive charges**

● **Blue** → **positive electrostatic potential**

- Electron-deficient regions
- **basic residues** (Lys, Arg, His+)
- attractive to **negative charges**

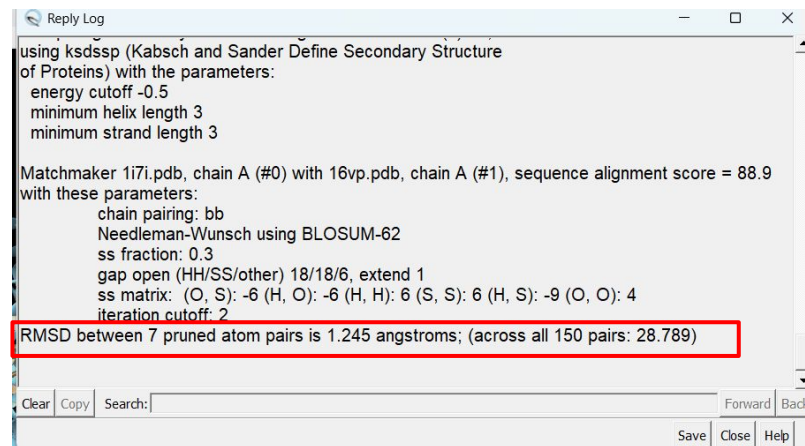
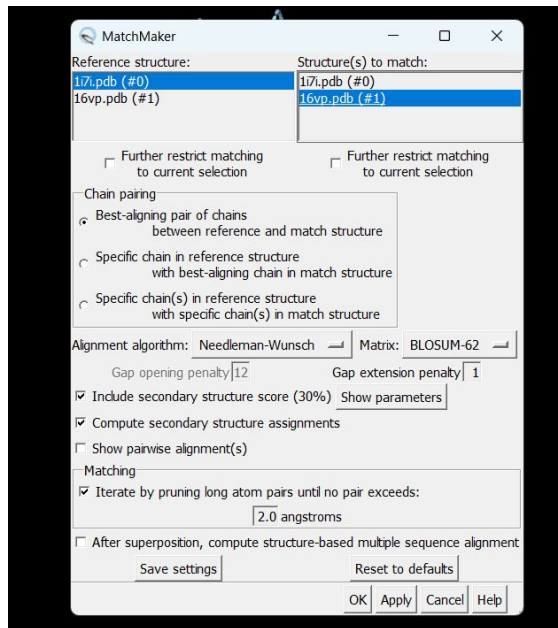
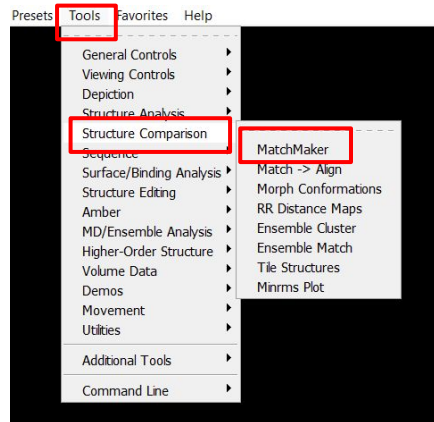
● **White** → **neutral or near zero potential**

- No strong electrostatic effect
- Nonpolar or uncharged regions

surface
coulombic -10 red 0 white 10 blue

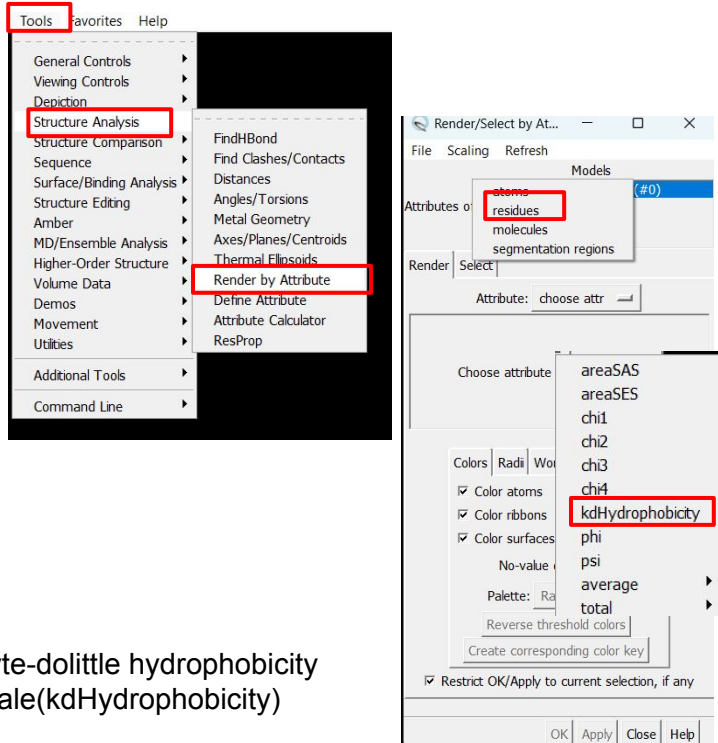
10. Matchmaker - Structure comparison

1. Open the structures you want to compare

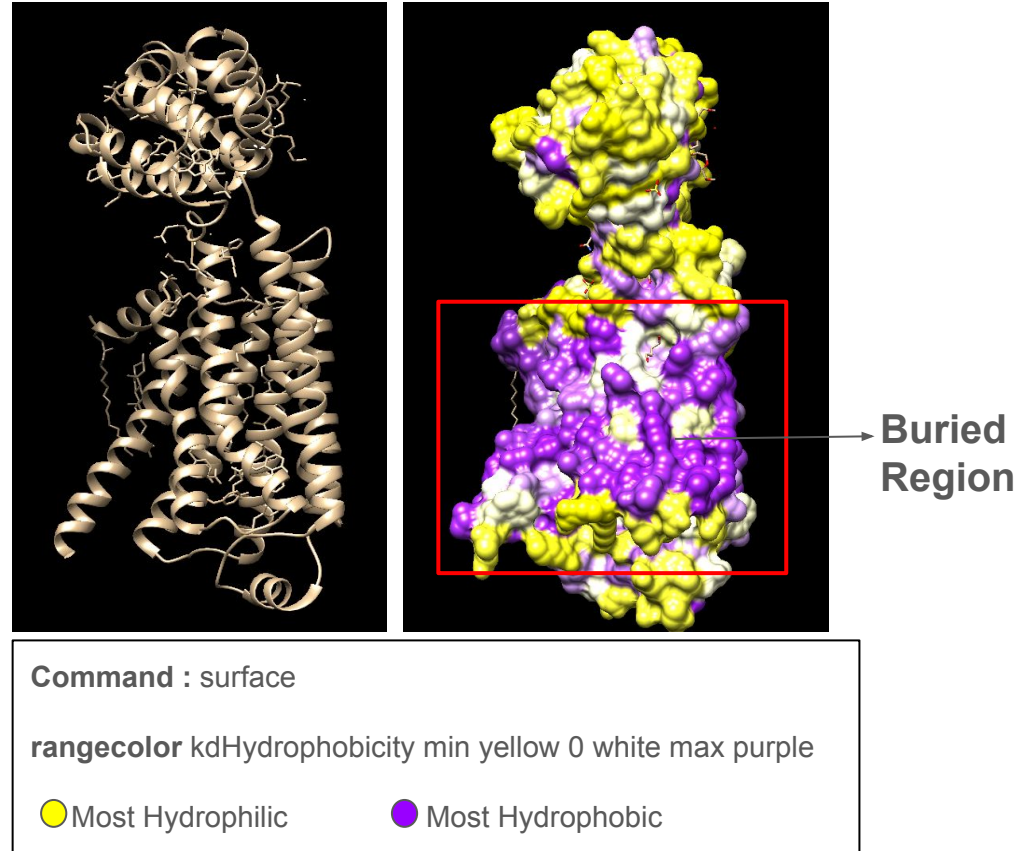


11. Hydrophobic and Hydrophilic regions of a membrane protein

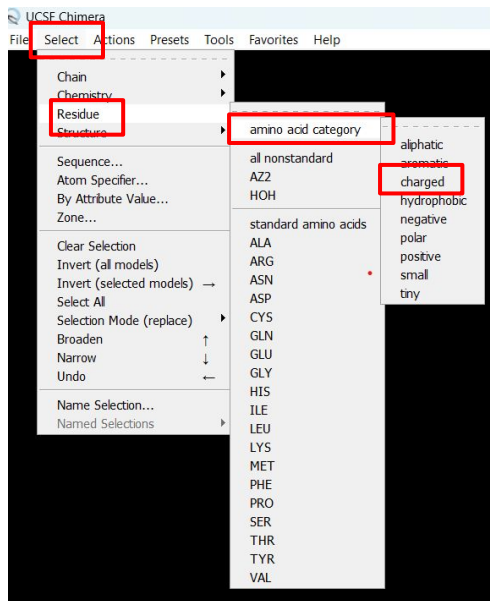
1. Open PDB **2RH1**
(**G protein-coupled receptor**
(**GPCR**) embedded in the cell
membrane.)



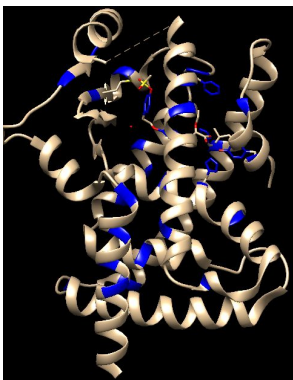
Kyte-doolittle hydrophobicity
scale(kdHydrophobicity)



12. Selecting residues on the basis of chemical properties



Charged residues are shown in pink color



Aromatic rings are shown in blue color

